UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of Computing
Semester 2
Session 2025/2026

SECJ3483 - SECTION 01
WEB TECHNOLOGY

Group Lab Activity: Security Investigation and Fix

Project Title : Secure Person BMI Full-Stack Web Application

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2. Security investigation summary

Test No	What I Tested	Expected Secure Behavior	Actual Result	Is It a Weakness?	Why?
1	Add BMI with invalid data	Backend should reject invalid data	Data is accepted and stored in the database.	Yes	Missing backend validation because frontend validation can be bypassed using ThunderClient.
2	Access another user's BMI record	Backend should deny access	The private BMI record data is returned.	Yes	Broken Object Level Authorization because data ownership is not verified after a user logs in.
3	Access staff route as normal user	Backend should deny access	The user is allowed to access and view the data.	Yes	Broken Function Level Authorization because there are no server-side role checks for staff features.
4	Access admin route as normal user	Backend should deny access	The user is allowed to access and view the data.	Yes	Broken Function Level Authorization because the route lacks server-side admin protection middleware.
5	Submit XSS payload in notes	Payload should display as text	The browser executes the script by showing the alert.	Yes	Stored Cross-Site Scripting because the frontend renders input unsafely using v-html.
6	Check API response	Password/hash should not be visible	The password value is visible in the payload.	Yes	Sensitive data exposure because the API blindly returns all columns including password_hash.
7	Try SQL Injection input	Input should be treated as data	The structural database query behaves unexpectedly.	Yes	SQL Injection because the application constructs database queries using unsafe string concatenation.
8	Try to update protected fields	Backend should reject or ignore protected fields	The fields (e.g., role, user_id) are accepted and modified	Yes	Mass assignment because the backend blindly executes updates on protected fields like user_id and role.

Test 1: Add BMI with invalid data

- Did the backend accept the data?

Yes, it saved the negative and empty values directly into the database.

- Did the frontend validation stop you?

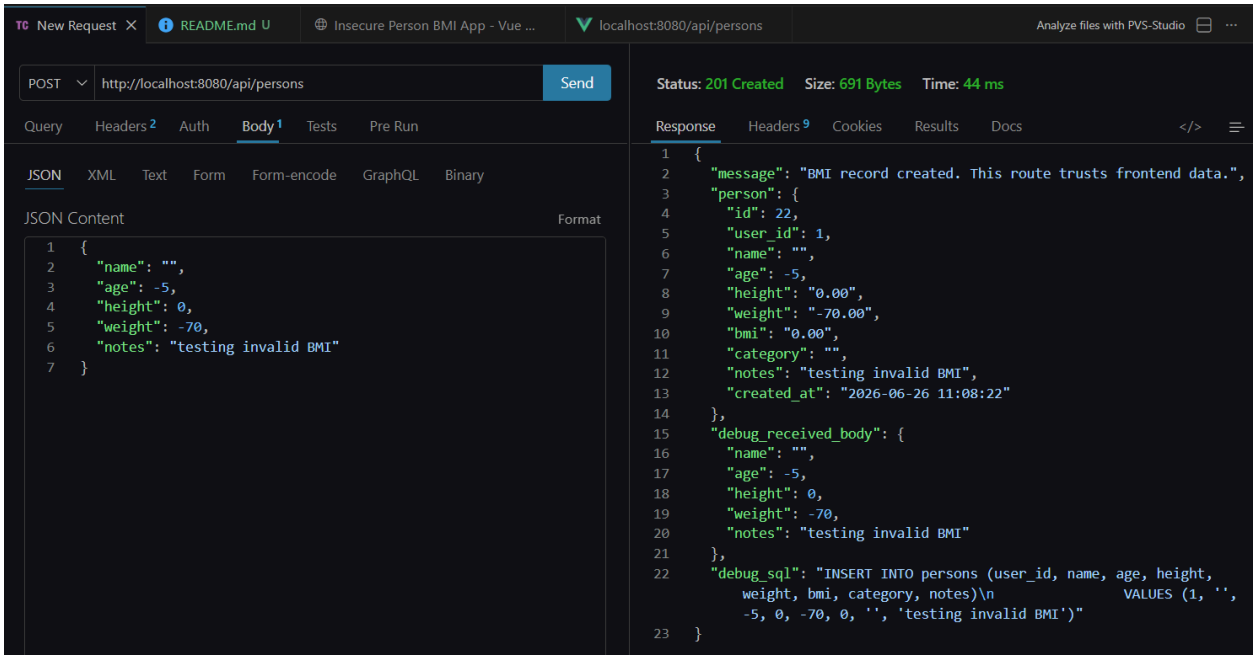
No, because I can bypass the frontend UI using Thunder Client.

- Can the same request be sent using Postman?

Yes, anyone can intercept and send raw requests to the API.

- Where should the validation be enforced?

Backend, because frontend validation can be bypassed.



Test 2: Access another user's BMI record

- Did the backend return another user's BMI record?

Yes, changing the ID in the URL leaked other people's data.

- Is login alone enough?

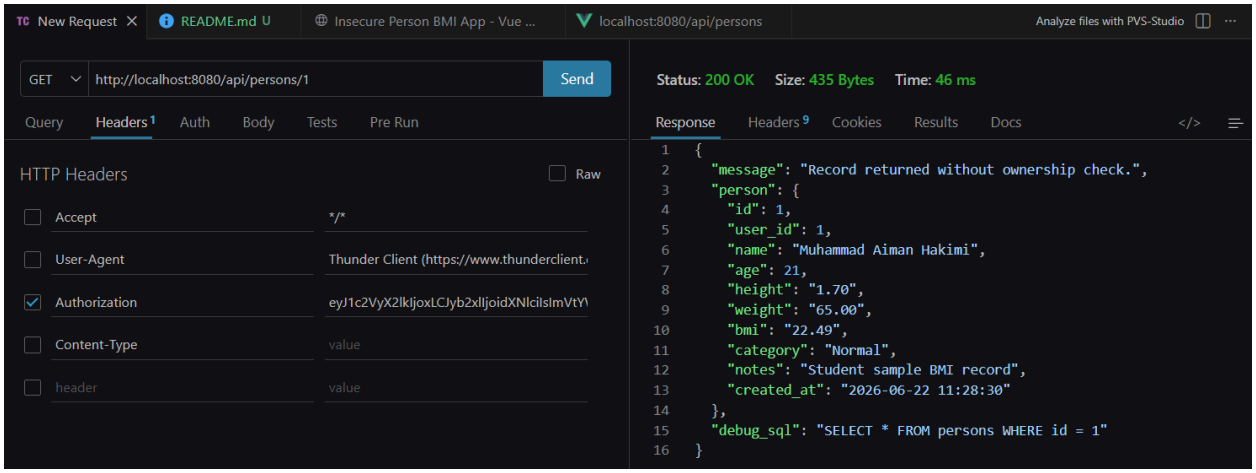
No, it is just authentication but no authorization.

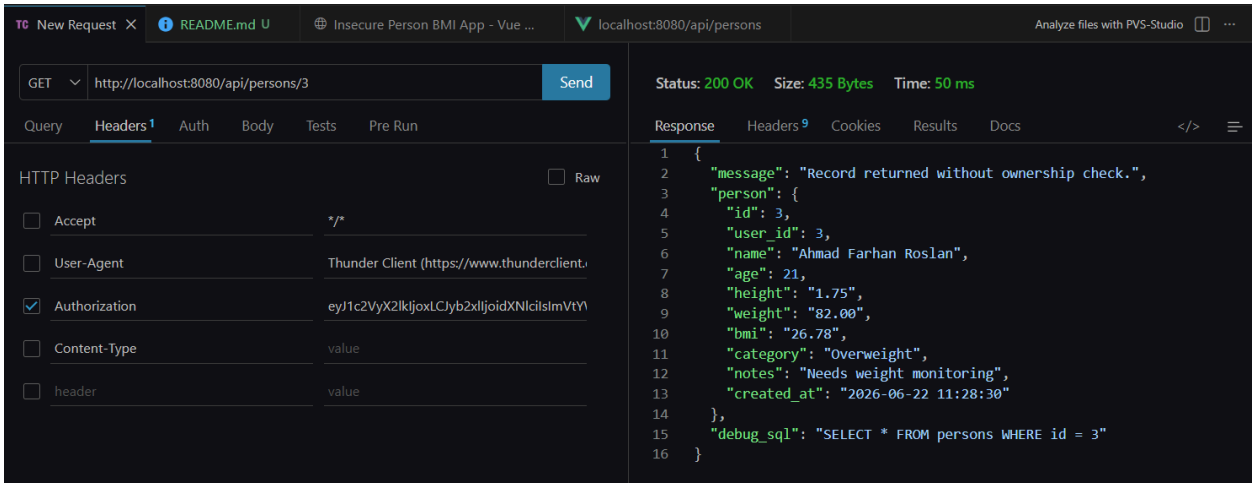
- What should the backend compare?

It must match the user_id from the JWT with the user_id on the database.

- Should staff and admin have different access?

Yes, staff and admin roles need to have different access to monitor all records.





Test 3: Access staff route as normal user

- Did the backend block the request?

No, a normal user token could access the restricted endpoints.

- Which roles should be allowed?

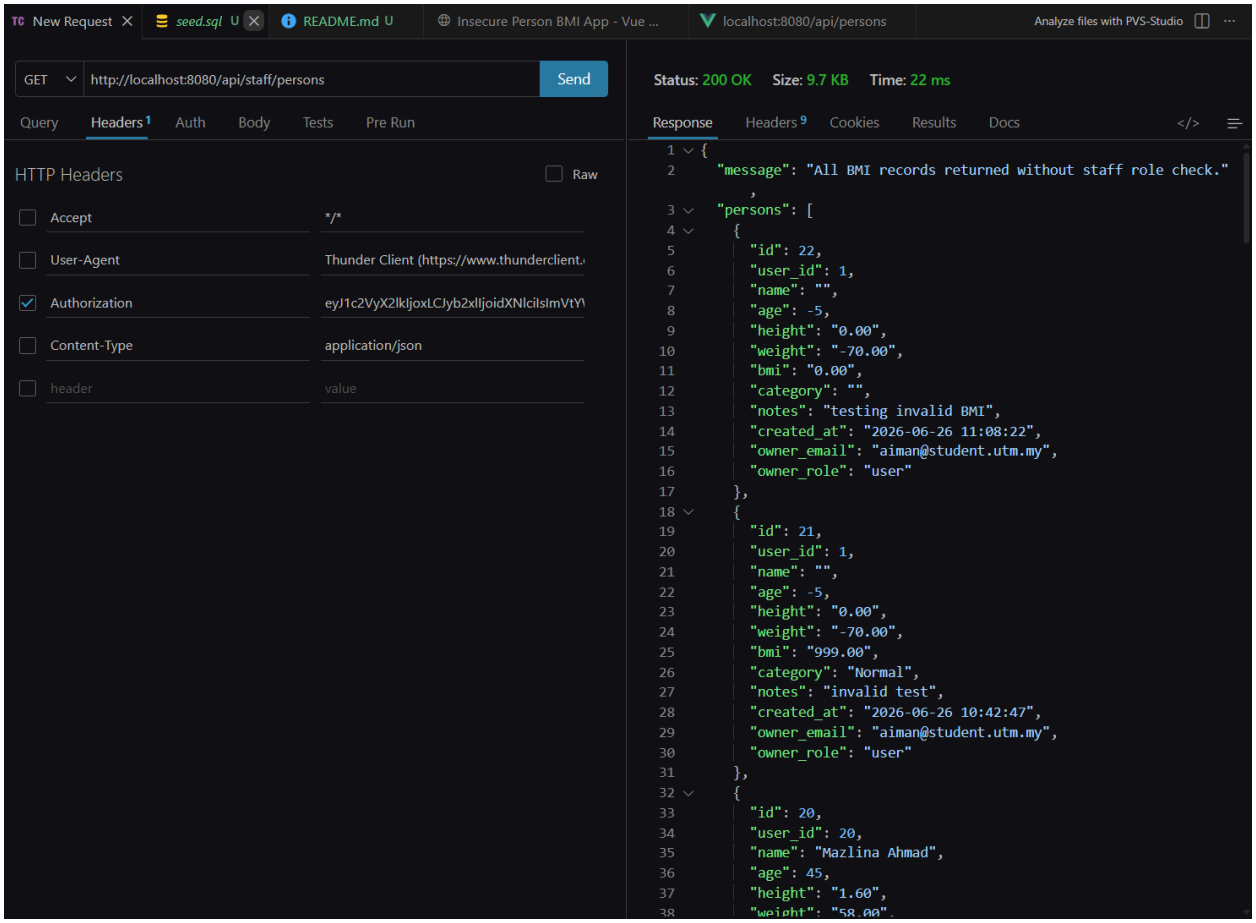
Only users with a role value of 'staff' or 'admin'.

- Should checking be done in frontend route guard only?

No, attackers don't use the frontend browser to make API requests.

- Where must the final decision happen?

In the backend routing middleware.



Test 4: Access admin route as normal user

- Did the backend accept user_id?

Yes.

- Did the backend accept role?

Yes.

- Did the backend accept bmi and category from the frontend?

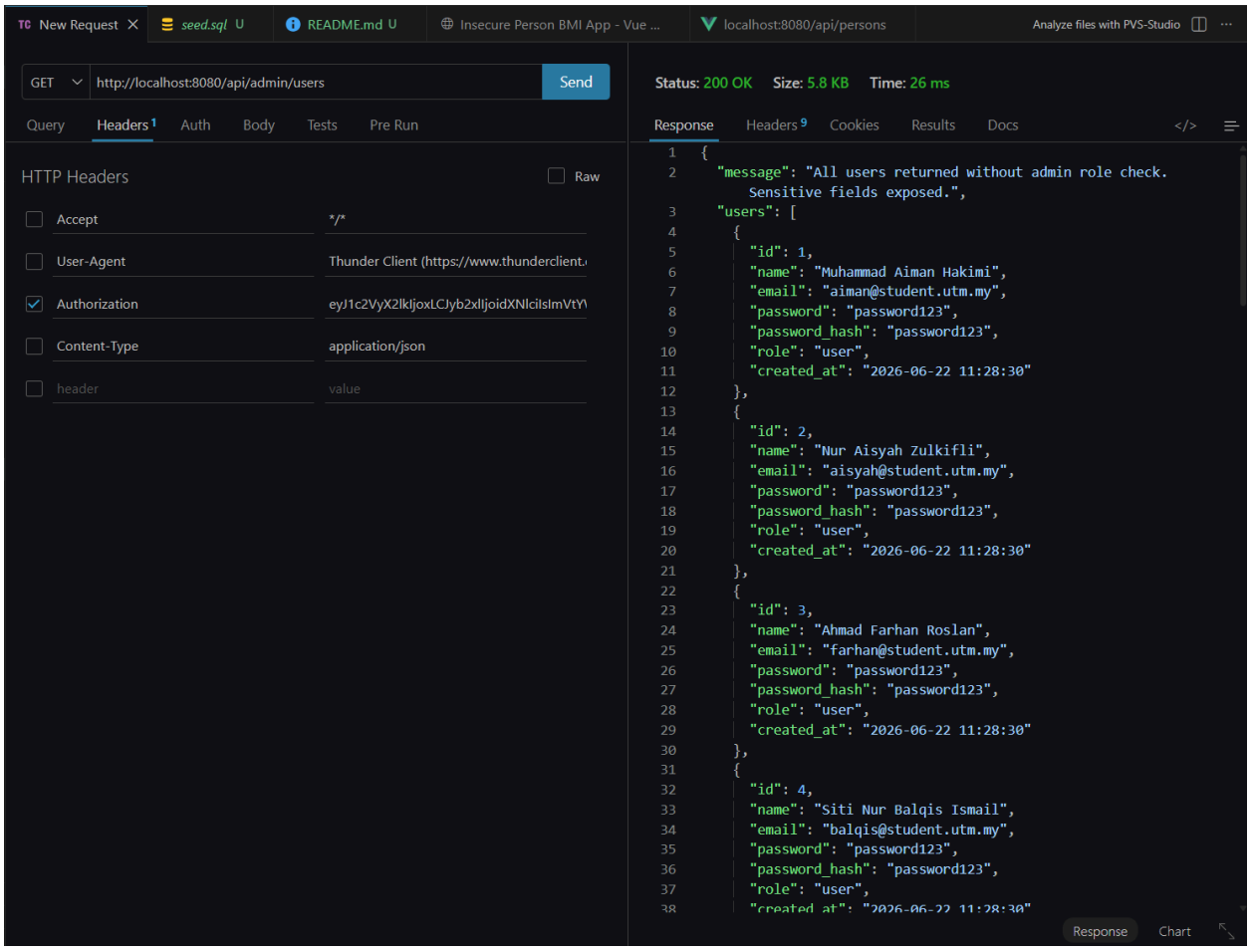
Yes, the database blindly updated them all.

- Which fields should be allowed?

Only name, age, height, weight, and notes.

- Which fields should be calculated or controlled by backend?

Structural fields like user_id and role, plus calculations like bmi and category.



Test 5: Submit XSS payload in notes

- Does the API return password_hash?

Yes, it was fully visible in the JSON response payload.

- Does the frontend really need this data?

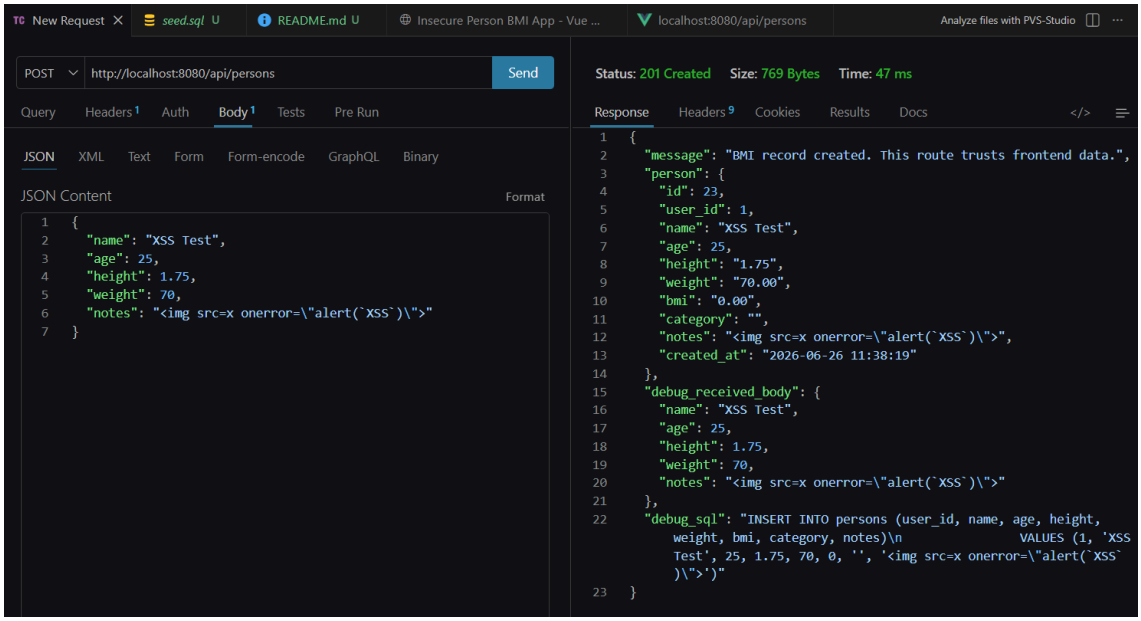
No, the client UI never needs to know password hashes.

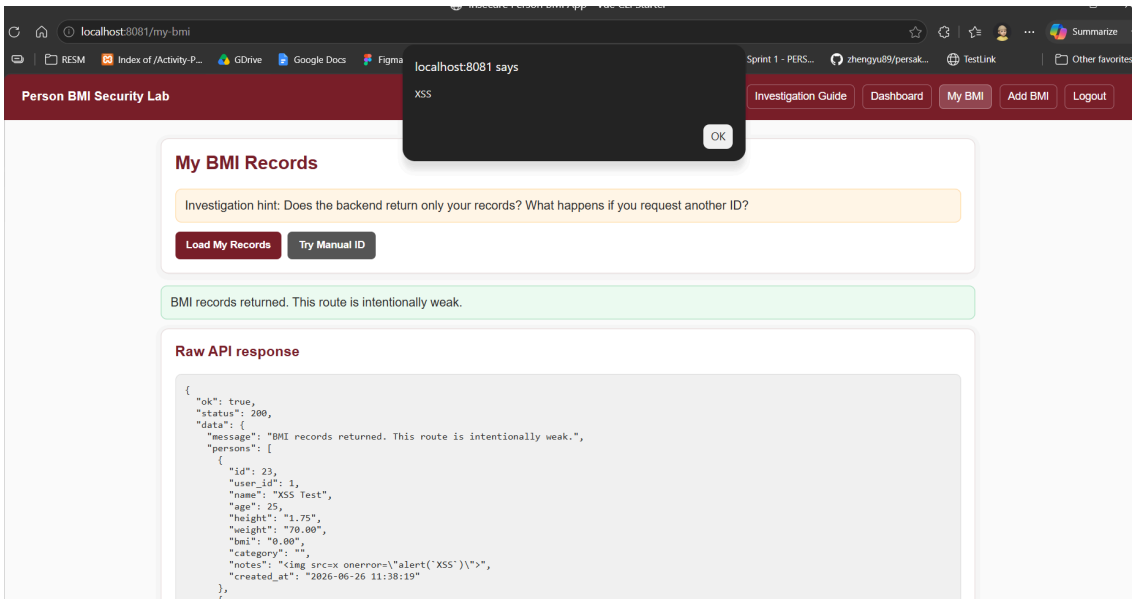
- What fields should be returned?

Only safe profile data: id, name, email, and role.

- How can SQL SELECT be improved?

Stop using SELECT * and explicitly write out the safe columns.





Test 6: Check API response

- Did the alert execute?

Yes, the script payload popped up in the browser window.

- Is the frontend using innerHTML or v-html?

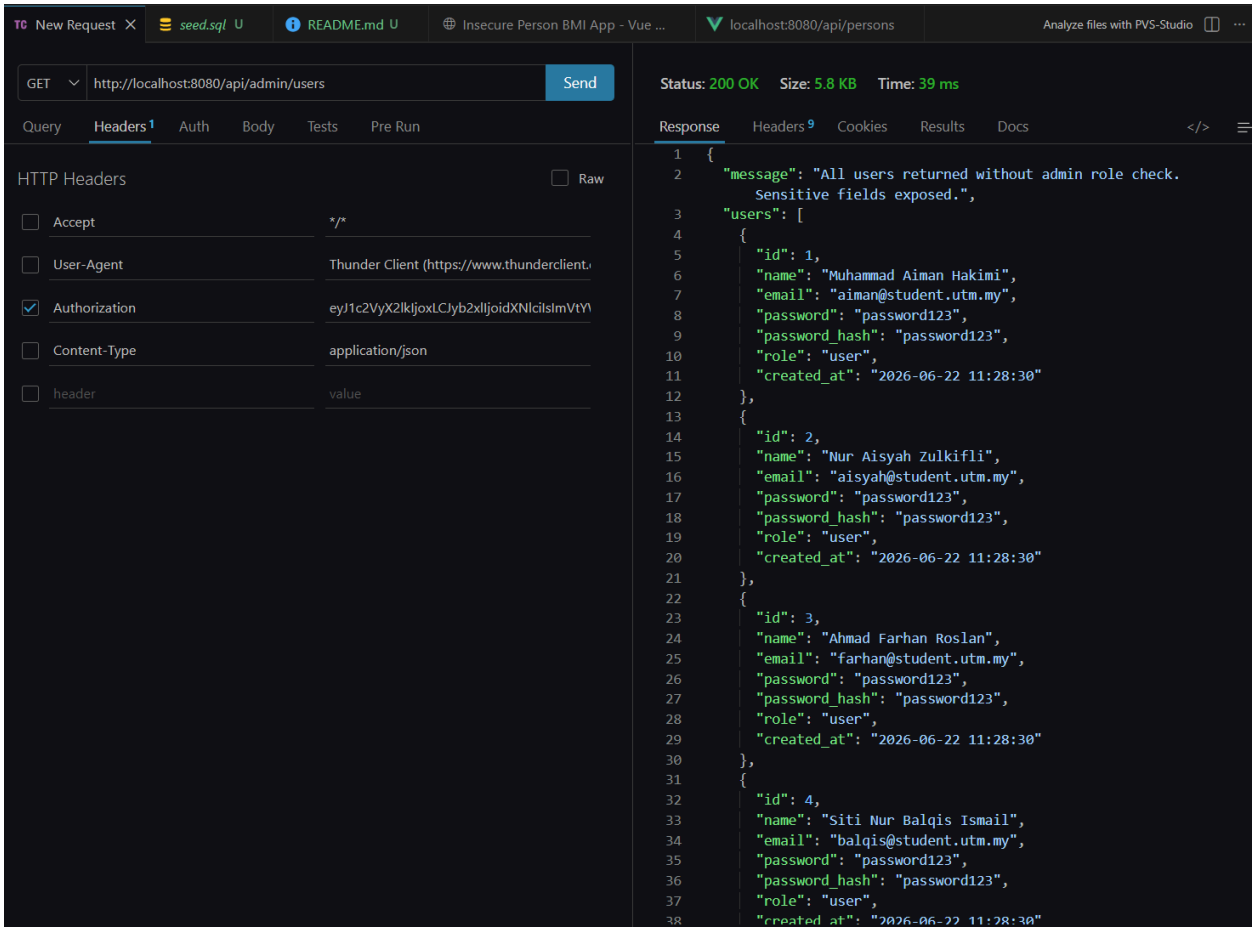
Yes, it used unsafe binding instead of text escaping.

- What is safer for displaying normal text?

Double curly braces {{ }} text interpolation or textContent.

- Should user-generated content be rendered as HTML?

No, never render raw user input as live HTML.



Test 7: Try SQL Injection input

- Did the SQL query behave unexpectedly?

Yes, inputting quotes broke out of the string and changed the logic.

- Was login bypassed?

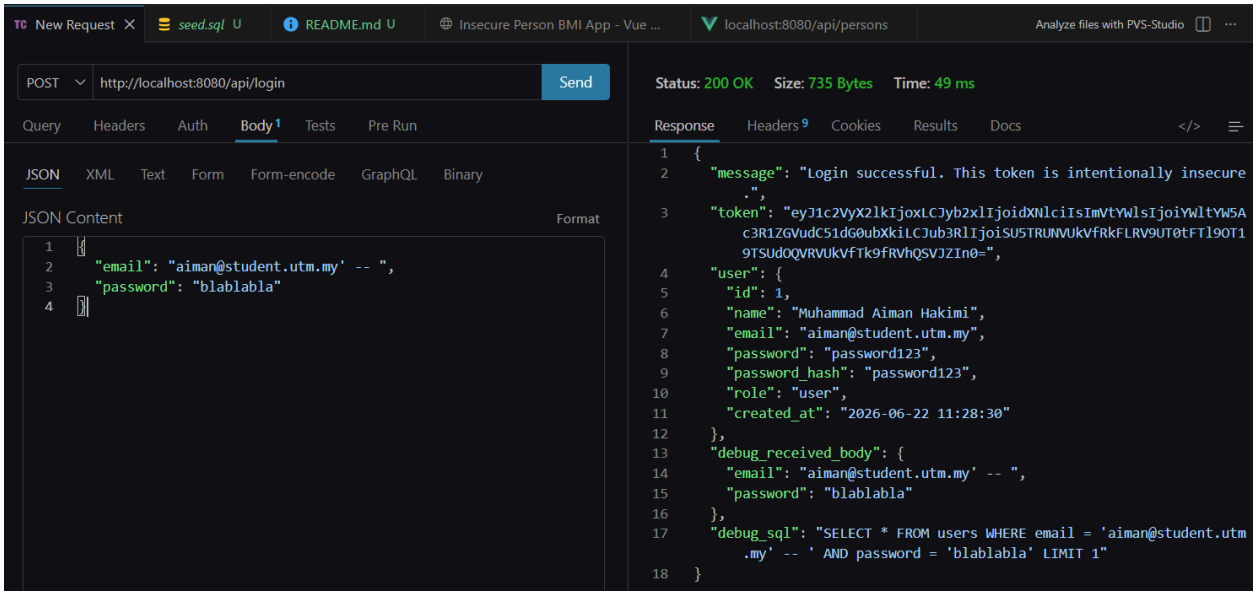
Yes, the payload successfully manipulated the query parameters.

- Was extra data returned?

Yes, it fetched records without validating real credentials.

- Is the SQL query using string concatenation?

Yes, it directly glued variables into the raw query string.



Test 8: Try to update protected fields